Statistics Note

1 Basic Knowledge

Probability Axioms: **1.** $\forall E \subseteq \Omega, \mathbb{P}(E) \geq 0.$ **2.** $\mathbb{P}(\Omega) = 1$ **3.** $\forall E_1, E_2 \subseteq \Omega, E_1 \cap E_2 = \emptyset,\ \mathbb{P}(E_1 \cup E_2) = \mathbb{P}(E_1) + \mathbb{P}(E_2)$

Marginal PDF: $f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)dy = \sum_y f_{X,Y}(x,y)$

Conditional PDF: $f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$

I.I.D. : Independent and identically distribution. ps: If $\mathbf{X} = (X_1, X_2, ..., X_n)\ i.i.d.\ f_X(x) = \prod_{i=1}^{n} f_{X_i}(x_i)$

Expected/Mean ($\mathbb{E}$): $\mathbb{E}(aX \pm bY \pm c) = a\mathbb{E}(X) \pm b\mathbb{E}(Y) \pm c$ If Independent: $\mathbb{E}(XY) = \mathbb{E}(X)\mathbb{E}(Y)$

Variance (Var): $Var[aX \pm bY \pm c] = a^2Var[X] + b^2Var[Y] \pm 2ab \cdot Cov(X,Y)$ If Independent: $Cov(X,Y) = 0$

· Other way to calculate variance: $Var[X] = \mathbb{E}[(X - \mathbb{E}(X))^2] = \mathbb{E}(X^2) - [\mathbb{E}(X)]^2$ $Std(X) = \sqrt{Var(X)}$

Independent: If $f_{X,Y}(x,y) = f_X(x)f_Y(y)$

Covariance and Correlation:

1. $Cov(X,Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = \sigma_X\sigma_Y R_{XY}$ $Cor(X,Y) = \frac{Cov(X,Y)}{\sqrt{Var(X)Var(Y)}} \in [-1,1]$
2. $Cov(X,X) = Var(X)$ $Cov(aX \pm b, cY \pm d) = ac \cdot Cov(X,Y)$

Central Limit Theorem (CLT): If $X_1, X_2, ..., X_n$ are i.i.d. :

· $\sum X_i \overset{apx.}{\sim} N(n\mu, n\sigma^2)$ $\frac{1}{n}\sum X_i \overset{apx.}{\sim} N(\mu, \frac{\sigma^2}{n})$ ps:$z = \frac{X-\mu}{\sigma^2}$

2 Estimation

2.1 Population, Sample and Estimator

Some Def: **Population** 总体 **Sample** 样本 **RVs** $(X_1, X_2, ..., X_n)$ 随机变量 **Census** 普查

Parameter (θ): 总体已知或未知的参数 **Representative**: 总体中每一个元素都有同等概率选中

Estimator (T_n): It's a function of Some RVs, used to estimate a Parameter. ps:Value called by the estimate.

Unbiased: If $\mathbb{E}(T_n) = \theta$ then we say that Estimator T_n is unbiased for θ. ps: Asymptotically unbiased:$\lim \mathbb{E}[T_n] = \theta$

Consistent: If $\lim \mathbb{E}(T_n) = \theta$ and $\lim Var(T_n) = 0$. We say that the estiamtor T_n is consistent.

· **1.** Unbiased better than Biased. **2.** 方差减少的快的比减少的慢的好.

2.2 Common Estimators

Sample Mean	Sample Var	S.D.	Cov(X,Y)	Cor(X,Y)	$\mathbb{P}(X \leq k)$
$\bar{X} = \frac{1}{n}\sum X_i$	$S^2 = \frac{1}{n-1}\sum(X_i - \bar{X})^2$	$S = \sqrt{S^2}$	$S_{xy} = \frac{1}{n-1}\sum(x_i - \bar{X})(Y_i - \bar{Y})$	$R_{xy} = \frac{S_{xy}}{S_xS_y}$	$P = \frac{1}{n}\sum \mathbb{1}_{[X_i \leq k]}$
μ 无偏，一致	σ^2 无偏，一致	σ 一致	σ_{xy} 无偏，一致	ρ 一致	π (p) 无偏，一致

Useful Formula: $Var[\bar{X}] = \sigma^2/n$ $S^2 = \frac{1}{n-1}(\sum X_i^2 - n\bar{X}^2)$

S.E. : $S.E.(\bar{X}) = \sqrt{Var[\bar{X}]} = \sqrt{\sigma^2/n}$ $\widehat{S.E}.(\bar{X}) = \sqrt{S^2/n}$

2.3 Method of Moments and Maximum Likelihood Estimate

Population Moments: $\mu_k = \mathbb{E}[X^k] = \sum_x x^k f_X(x|\theta) = \int_{\mathbb{R}} x^k f_X(x|\theta)dx$

Sample Moments: $\overline{X^k} = \frac{1}{n}\sum X_i^k$

Method of Moments: If $\theta = f(\mu_1, \mu_2, ..., \mu_n)$. Then $\hat{\theta} = f(\overline{X}, \overline{X^2}, ..., \overline{X^n})$ ps: Always Consistent.

(Log-)Likelihood Function: $L(\theta|\mathbf{x}) = \prod f(x_i;\theta)$ $\ell(\theta|\mathbf{x}) = ln[L(\theta|\mathbf{x})]$

Maximum likelihood Estimate (MLE): $\hat{\theta} = ArgMax[L(\theta|\mathbf{x})] = ArgMax[\ell(\theta|\mathbf{x})]$ ps: Always Consistent.

· Condition: x_i need i.i.d. ArgMax 有条件约束最大值 通过求导 =0 得到 $\hat{\theta}$ (不是必须,e.g.$X \sim U(a,b)$)

2.4 Confidence Interval

Def of Confidence Interval: $100(1-\alpha)\%$ Confidence Interval for θ is $[l(X), u(X)]$, $\mathbb{P}(l(X) < \theta < u(X)) = 1 - \alpha$.

· Interval 越小代表 uncertainty 越低. ↑%,wider CI CI 越小越好 ps: 我们只考虑 (The central confidence interval) 枢轴变量.

How to choose	Known (σ/μ)	Unknown (σ/μ)
Find μ	By CLT $\frac{\bar{X}-\mu}{\sqrt{\sigma^2/n}} \sim N(0,1)$ n 很大时，$\sigma^2 = S^2$ CI: $(\bar{X} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \bar{X} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}})$	$\frac{\bar{X}-\mu}{\sqrt{S^2/n}} \sim t_{n-1}$ CI: $(\bar{X} - t_{n-1;\alpha/2}\frac{S}{\sqrt{n}}, \bar{X} + t_{n-1;\alpha/2}\frac{S}{\sqrt{n}})$
Find σ^2	$\frac{1}{\sigma^2}\sum(X_i - \mu)^2 \sim \chi_n^2$ CI: $(\frac{\sum(X_i-\mu)^2}{\chi^2_{n;\alpha/2}}, \frac{\sum(X_i-\mu)^2}{\chi^2_{n;1-\alpha/2}})$	$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}$ CI: $(\frac{(n-1)S^2}{\chi^2_{n-1;\alpha/2}}, \frac{(n-1)S^2}{\chi^2_{n-1;1-\alpha/2}})$

· We need $X_1, X_2, ..., X_n \sim N(\mu, \sigma^2)$ ps:$z_p := \mathbb{P}(Z \geq z_p) = p$

3 Hypothesis Testing

3.1 Null, Alternative Testing, Power, P-value and Significance Level

Null Hypothesis (H_0): Claim about some propert of the population. e.g. ($H_0 = \theta$)

Alternative Hypothesis (H_1): an assertion about how population differ to contradict H_0. (Two or One Side)

Type	Type I error	Type II error
Def	$\mathbb{P}(Rej\ H_0 \mid H_0\ is\ T)$	$\mathbb{P}(Fail\ Rej\ H_0 \mid H_0\ is\ F)$

-	H_0 is T	H_0 is F
Fail to Rej H_0	Right	Type II (β) 与 size 有关
Rej H_0	Type I (α) 与 size 无关	Right (Power)

Significance Level: $\alpha = \mathbb{P}(Rej\ H_0 \mid H_0\ is\ T)$

Power of Test: $\mathbb{P}(\text{Rej } H_0 \mid H_0 \text{ is F}) = 1 - \beta = \mathbb{P}(Z \in C \mid \mu = \mu^*)$

· 1.Sample size ↑, Power ↑; 2.σ^2 ↓, Power ↑; 3.Effect size $|\theta_0 - \theta^*|$ ↑, Power ↑; 4.α↑, Power ↑.

P-value: 1.One-side: $p = \mathbb{P}(T \geq t; T \sim H_0)$ 2.Two-sides: $p = 2\min\{\mathbb{P}(T \geq t; T \sim H_0), \mathbb{P}(T \leq t; T \sim H_0)\}$

3.2 Test Statistics and Decision Rule

One Sample		Two Sample		
Find μ		Find μ		Find σ^2
Z-Test (σ^2 known) $Z = \frac{\bar{X}-\mu_0}{\sqrt{\sigma^2/n}} \sim N(0,1)$	t-Test (σ^2 unknown) $T = \frac{\bar{X}-\mu_0}{\sqrt{S^2/n}} \sim t_{n-1}$	Paired t-Test (X,Y same Unit) $D_i = Y_i - X_i \sim N(\mu_D, \sigma_D^2)$ $T = \frac{\bar{D}-\mu_D}{\sqrt{S^2/n}} \sim t_{n-1}$	Two-Sample t-Test (独立) $T = \frac{(\bar{X}-\bar{Y})-\Delta_0}{\sqrt{\sigma^2(\frac{1}{m}+\frac{1}{n})}} \sim N(0,1)$ 未知:S_p^2;$T \sim t_{m+n-2}$	F-Test $F = \frac{S_X^2}{S_Y^2}$ $\sim F_{m-1,n-1}$

· 对于 Two-Sample t-Test 我们要求 X,Y σ^2 一致 ps:$H_0: \mu_D = \mu_X - \mu_Y$; $H_0: \Delta_0 = \mu_x - \mu_Y$; $X: m, Y: n$

Pooled Sample Variance: (Two sample Variance) $S_p^2 = \frac{(m-1)S_X^2 + (n-1)S_Y^2}{m+n-2}$ ps:Unbiased

3.3 Conclusion and Assessing Normality

Conclusion:

1. Rejecting H_0 in favour of H_1. (If it is in Critical Region C).
2. Failing to reject the H_0. (If it is not in Critical Region C)

The Expirical Cumulative Distribution Function: Let $x_1 \leq x_2 \leq \cdots \leq x_n$ be observed sample.

· ECDF:$\hat{F}_X(x) = \frac{1}{n}\sum_{i=1}^n \mathbb{1}_{[x_i < x]}$ ps:Unbiased for $F(x)$ CDF.

QQ-Plot: (z_i, x_i) where $z_i = \Phi^{-1}(\frac{i-0.5}{n})$ 如果中心点是 (0,0) 为 standard normal 否则为 non-standard normal

· ps: 如果是正态分布，图像大致沿一条直线分布，两端相对平滑。(两端同时轻微向上或向下也算)

· ps:negative skew：两端都在线下，反之 positive skew; light tails: 两端取向横向零，否则 heavy tails.

4 Linear Regression

4.1 Simple Linear Regression Model

Method of Least-Squares: Consider the model $\mathbb{E}[Y] = \alpha + \beta x$ or model $Y_i = \alpha + \beta x_i + \epsilon_i$ where $\mathbb{E}[\epsilon_i] = 0, Var[\epsilon_i] = \sigma^2$.

-	Estimator (Unbiased)	Var / S.E. For $\widehat{Var}, S_e^2 = \sigma^2$
α	$\hat{\alpha} = \bar{Y} - \hat{\beta}\bar{x}$	$Var[\hat{\alpha}] = \sigma^2(\frac{1}{n} + \frac{\bar{x}^2}{(n-1)S_x^2})$ (Unbiased)
β	$\hat{\beta} = \frac{\sum(x_i-\bar{x})(Y_i-\bar{Y})}{\sum(x_i-\bar{x})^2} = \frac{S_{xy}}{S_x^2} = r_{x,y} \cdot \frac{S_y}{S_x}$	$Var[\hat{\beta}] = \frac{\sigma^2}{(n-1)S_x^2}$ (Unbiased)
Other	$r_{x,y} = \frac{S_{xy}}{S_x S_y}$ **Fitted regression line**:$\hat{y}_i = \hat{\alpha} + \hat{\beta}x_i$ **Residuals**:$e_i = y_i - \hat{y}_i$ ps:Var / S.E. are Unbiased.	
Other	**Residual Sum of Squares (Q(α,β))**: $\sum(Y_i - \hat{Y}_i)^2$ **Residual Variance**: $S_e^2 = \frac{1}{n-2}\sum(y_i - \hat{y}_i)^2$ (Unbiased for σ^2)	
Other	Another formula for $S_e^2 = \frac{n-1}{n-2}(S_y^2 - \hat{\beta}^2 S_x^2)$	In R studio: $H_0: \alpha, \beta = 0$

4.2 Normal Linear Regression Model

-	Distribution	Marginal Distribution	Confidence Interval
α	$\hat{\alpha} \sim N(\alpha, \sigma^2[\frac{1}{n} + \frac{\bar{x}^2}{(n-1)S_x^2}])$	$\frac{\hat{\alpha}-\alpha}{\sqrt{S_e^2(\frac{1}{n}+\frac{\bar{x}^2}{(n-1)S_x^2})}} \sim t_{n-2}$	$\hat{\alpha} \pm t_{n-2;\alpha/2}\sqrt{S_e^2(\frac{1}{n} + \frac{\bar{x}^2}{(n-1)s_x^2})}$
β	$\hat{\beta} \sim N(\beta, \frac{\sigma^2}{(n-1)S_x^2})$	$\frac{\hat{\beta}-\beta}{\sqrt{\frac{S_e^2}{(n-1)S_x^2}}} \sim t_{n-2}$	$\hat{\beta} \pm t_{n-2;\alpha/2}\sqrt{\frac{S_e^2}{(n-1)S_x^2}}$

· ps:$\frac{(n-2)S_e^2}{\sigma^2} \sim \chi_{n-2}^2$ We assume that $Y_i \overset{indep.}{\sim} N(\alpha + \beta x_i, \sigma^2)$ or $Y_i = \alpha + \beta x_i + \epsilon_i, \epsilon \overset{i.i.d.}{\sim} N(0, \sigma^2)$:

Caution(We don't use): Some author use $S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2, and\ S_{xy} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$ and $\hat{\beta} = \frac{S_{xy}}{S_{xx}}$

Theorem: The MLE for α, β = Least-Squares, If $Y_i \overset{indep.}{\sim} N(\alpha + \beta x_i, \sigma^2)$ or $Y_i = \alpha + \beta x_i + \epsilon_i, \epsilon \overset{i.i.d.}{\sim} N(0, \sigma^2)$

· By using MLE, $\widehat{\sigma^2} = \frac{1}{n}\sum_{i=1}^n (y_i - \hat{\alpha} - \hat{\beta}x_i)^2 = \frac{n-2}{n}S_e^2 (Biased)$

4.3 Confidence Interval for $\mathbb{E}[Y_0]$ and Prediction for Y_0

Confidence Interval for $\mathbb{E}[Y_0]$: Regression Equ $\mathbb{E}[Y_0] = \alpha + \beta x_0$, $\mathbb{E}[Y_0]$ is fixed but unknown.

· **CI**: $\hat{\alpha} + \hat{\beta}x_0 \pm t_{n-2;\alpha/2}\sqrt{S_e^2(\frac{1}{n} + \frac{(x_0-\bar{x})^2}{(n-1)S_x^2})}$ ps: 对于群体. $Var[\hat{\mathbb{E}}(Y_0)] = \sigma^2[\frac{1}{n} + \frac{(x_0-\bar{x})^2}{(n-1)S_x^2}]$ $\frac{\hat{\mathbb{E}}[Y_0] - \mathbb{E}[Y_0]}{\sqrt{S_e^2(\frac{1}{n} + \frac{(x_0-\bar{x})^2}{(n-1)s_x^2})}} \sim t_{n-2}$

Prediction Interval for Y_0: Regression Equ $Y_0 = \alpha + \beta x_0 + \epsilon_0$, independent error $\epsilon_0 \sim N(0, \sigma^2)$

· **PI**: $\hat{\alpha}+\hat{\beta}x_0 \pm t_{n-2;\alpha/2}\sqrt{S_e^2(1+\frac{1}{n}+\frac{(x_0-\bar{x})^2}{(n-1)S_x^2})}$ ps: 对于个体. $Var[\hat{Y}_0]=\sigma^2[1+\frac{1}{n}+\frac{(x_0-\bar{x})^2}{(n-1)s_x^2}]$

Difference between them: Consider fitted regression line $\hat{\alpha}+\hat{\beta}x_0$ then:

1. $\hat{\mathbb{E}}[Y_0]$: The estimate of the regression line at new explantory value.
2. $\mathbb{E}[\hat{Y}_0]$: The expected value of some unobserved RVs.
3. PI interval > CI interval. (Uncertainty from single future observation.) ps: 离 sample mean x 越远，intervals↑.

4.4 Check Assumptions

Assumption for Normal Linear Regression Model: $Y_i \overset{i.i.d.}{\sim} N(\alpha+\beta x_i,\sigma^2)$

Decomposition of Variability: $\sum_{i=1}^n(y_i-\bar{y})^2$ $_{(Total\ Varia)}$ = $\sum_{i=1}^n(y_i-\hat{y}_i)^2$ $_{(Unexplained\ Varia)}$ + $\sum_{i=1}^n(\hat{y}_i-\bar{y})^2$ $_{(Explained\ Varia)}$

Def of R-Squared: R-Squared statistic describes the proportion of variability explained by the fitted regresison line.

Multiple R-Squared: $R^2=\frac{\sum(\hat{y}_i-\bar{y})^2}{\sum(y_i-\bar{y})^2}=1-\frac{\sum(y_i-\hat{y}_i)^2}{\sum(y_i-\bar{y})^2}\in[0,1]$ ps:R 越大 better predicts the data.

· 代表：**1.** The linear model explains $\{R^2\}$ of the Variability within the $\{y\}$ data.

2. $\{1-R\}$ of Variability of the $\{y\}$ data remains unexplained.

· 格式: As R^2 for the {Model 1} {?%} is larger (smaller) than {Model 2} {?%}, so {Model 1} is preferred as it expains more of the response variability.

Reliability of the prediction: 计算 $k=\frac{|x_0-\bar{x}|}{s_x}$ ps: 一般来说小于 1.96 以上才说好的 reliability.

· Linear Regression 从 $\bar{x}$ 往左右，reliable 逐渐下降.

· 格式：The value of the explantory variable is about {k} S.E. above (below) the sample mean $\bar{x}$. So point x_0 can (cannot) be considered to be within the reasonable range of the explantory variables and not (will) be considered as a point of extrapolation. So it's (not) reliable.

Graphical Assessment

1. Residuals vs Fitted (Check linear assumption): $(\hat{y}_i,e_i)$, $e_i=y_i-\hat{y}_i$, $\hat{y}_i=\hat{\alpha}+\hat{\beta}x_i$ 在 0 附近，越靠 0 越好。
   ps: 点在整张图均匀分布 →constant variance (the range of the scatter of points is consistent over the range of fitted values)
2. Scale-Location (Check constant variance assumption): $(\hat{y}_i,\sqrt{|e_i^*|})$, $e_i^*=\frac{e_i}{s_e}$ 越靠近 0.82 越好.(around a constant value)
3. Normal QQ-Plot (Check Normal Assumption): (z_i,e_i^*) , $z_i=\Phi^{-1}(\frac{i-0.5}{n})$ 判断方法具体见 3.3

5 Multiple Regression and Analysis of Variance

5.1 Extending Least-Squares Estimation

Definitions

$Y_i \overset{i.i.d.}{\sim} N(\beta_1x_{i,1}+\beta_2x_{i,2}+\cdots+\beta_px_{i,p},\sigma^2)$ for $i=1,...,n$

$$For:\begin{pmatrix}Y_1\\Y_2\\\vdots\\Y_n\end{pmatrix}=\begin{pmatrix}x_{1,1}&\cdots&x_{1,p}\\x_{2,1}&\cdots&x_{2,p}\\\vdots&\ddots&\vdots\\x_{n,1}&\cdots&x_{n,p}\end{pmatrix}\begin{pmatrix}\beta_1\\\beta_2\\\vdots\\\beta_n\end{pmatrix}+\begin{pmatrix}\epsilon_1\\\epsilon_2\\\vdots\\\epsilon_n\end{pmatrix}$$

we write as: $\underline{Y}=\boldsymbol{X}\underline{\beta}+\underline{\epsilon}$ **截距也算一个 p**

· ps: $\underline{Y}$:Response Random vector; $\underline{\beta}$:Parameter vector;

· ps: $\underline{\epsilon}$:Error Random vector; $\boldsymbol{X}$:Design matrix;

Sum of Squares:$Q(\underline{\beta})=\sum_{i=1}^n(y_i-\mathbb{E}[Y_i])^2=(\underline{y}-\boldsymbol{X}\underline{\beta})^T(\underline{y}-\boldsymbol{X}\underline{\beta})$

Estimates and some Formulas

Least-Squares Estimates: $\underline{\hat{\beta}}=(\boldsymbol{X}^T\boldsymbol{X})^{-1}\boldsymbol{X}^T\underline{y}$

· ps: Cooresponding MLE estimates for $\underline{\beta}$ are the same.

$\mathbf{S_e^2}=\frac{1}{n-p}\sum_{i=1}^n[Y_i-\hat{Y}_i]^2=\frac{1}{n-p}(\underline{y}-\boldsymbol{X}\underline{\hat{\beta}})^T(\underline{y}-\boldsymbol{X}\underline{\hat{\beta}})$

· ps: $\frac{(n-p)S_e^2}{\sigma^2}\sim\chi^2_{n-p}$ (Residual Variance Estimator)

Marginal Distribution: $\frac{\hat{\beta}_j-\beta_j}{S.E.(\hat{\beta}_j)}\sim t_{n-p}$ for $j=1,2,...,p$

CI for $\hat{\beta}_j$: $\hat{\beta}_j\pm t_{n-p;\alpha/2}\widehat{S.E.}(\hat{\beta}_j)$ for $j=1,2,...,p$

· ps: we don't learn the S.E. formula for β

Adjusted-R^2: $R^2_{adj.}=1-(1-R^2)\frac{n-1}{n-p}$

· 多个 x 时用 adj, 单个 x 用 multiple

ps: In R studio: $H_0:\beta_i=0$

5.2 ANOVA F-test for model comparison

Full Model: $Y_i \overset{i.i.d.}{\sim} N(\beta_1x_{i,1}+\cdots+\beta_{p_0}x_{i,p_0}+\cdots+\beta_{p_1}x_{i,p_1},\sigma^2)$ ps: 假设 $1-p_0$ 是有关的,p_0-p_1 是无关的.

Sub-Model: $Y_i \overset{i.i.d.}{\sim} N(\beta_1x_{i,1}+\cdots+\beta_{p_0}x_{i,p_0},\sigma^2)$ ps:$\frac{RSS_0}{\sigma^2}\sim\chi^2_{n-p_0}$, $\frac{RSS_1}{\sigma^2}\sim\chi^2_{n-p_1}$ and $\frac{RSS_0-RSS_1}{\sigma^2}\sim\chi^2_{p_1-p_0}$

RSS (Residual sum of squares): $RSS_0=(n-p_0)S_0^2$, $RSS_1=(n-p_1)S_1^2$ ps:(S_1 / S_0: residual var estiamtor for full model / sub model)

Def of ANOVA(Analysis of Variance) F-test: Consider model $\underline{Y}=X\underline{\beta}+\underline{\epsilon}$ where $\epsilon_i\overset{i.i.d}{\sim}N(0,\sigma^2)$, $\underline{\beta}=(\beta_1,...,\beta_{p_0},...,\beta_{p_1})$

1. **Hypothesis**: H_0: $\beta_{p_0+1}=\cdots=\beta_{p_1}=0$ vs H_1: $\exists j\in\{p_{0+1},...,p_1\}$ s.t. $\beta_j\neq0$
2. **F-test Statistic**: $F=\frac{(RSS_0-RSS_1)/(p_1-p_0)}{RSS_1/(n-p_1)}\sim F_{p_1-p_0,n-p_1}$, under H_0 (ps: $F=\frac{R^2}{1-R^2}\cdot\frac{n-(p_1-1)-1}{(p_1-1)}$ 对于 R studio: 除了截距其他为 0)
   · ps: 等价于：$\frac{RSS_0}{\sigma^2}\sim\chi^2_{n-p_0}$, $\frac{RSS_0-RSS_1}{\sigma^2}\sim\chi^2_{p_1-p_0}$ and RSS_0-RSS_1 is independent of RSS_1 ; under H_0
3. **Changed Hypothesis**: $H_0:F=0$ vs $H_1:F>0$
4. **Critical Region**: $C=[F_{p_1-p_0,n-p_1;\alpha},+\infty)$

5.3 One-way ANOVA

Target: 比较单类多组 (≥3) 之间的均值是否有显著性差异。 $H_0: \mu_1 = \mu_2 = \cdots = \mu_k$ $H_1: \mu_1, ..., \mu_k$ are not all equal.

· 这里的组可以理解为 sample，如果组为 2，那么就是前面的 two-sample t-test.

Definitions: Response Variables (k 组，每组 n_k 个数据): $\underline{Y} = (Y_{1,1}, ..., Y_{n_1,1}, ..., Y_{1,k}, ..., Y_{n_k,k})^T$ where $Y_{i,j} \overset{i.i.d.}{\sim} N(\mu_i, \sigma^2)$

Design matrix: $\mathbf{X} = (\underline{x}_1, \underline{x}_2, ..., \underline{x}_k)$ Parameters: $\underline{\beta} = \underline{\mu} = (\mu_1, ..., \mu_k)^T$ $\mathbb{E}[\underline{Y}] = \mathbf{X}\underline{\mu}$

OR: $\tilde{\mathbf{X}} = (\underline{1}, \underline{x}_2, ..., \underline{x}_k)$ and $\underline{\beta} = (\mu_1, \beta_2 ..., \beta_k)^T$. $\beta_j = \mu_j - \mu_1$

Estimators: $\hat{\mu}_j = \bar{y}_j$(Group Sample Mean)$= \frac{1}{n_j}\sum_{i=1}^{n_j} y_{i,j}$ Residual Variance: $S_e^2 = \frac{1}{n-k}\sum_{j=1}^{k}\sum_{i=1}^{n_j}(y_{i,j} - \hat{\mu}_j)^2$ Overall Sample Mean: $\frac{1}{n}\sum_{j=1}^{k}\sum_{i=1}^{n_j} Y_{i,j}$

$RSS_y = (n-1)S_y^2 = \sum_{j=1}^{k}\sum_{i=1}^{n_j}(Y_{i,j} - \bar{Y})^2$ $RSS_b = (k-1)S_b^2 = \sum_{j=1}^{k} n_j(\bar{Y}_j - \bar{Y})^2$ $RSS_e = \sum_{j=1}^{k}(n_j - 1)S_j^2 = \sum_{j=1}^{k}\sum_{i=1}^{n_j}(Y_{i,j} - \bar{Y}_j)^2$

ps: $RSS_y = RSS_b + RSS_e$ RSS_e error residual sum of squares *within* groups. $RSS_b \sim$ *between* groups.

F-test statistic: $F = \frac{(RSS_y - RSS_e)/(k-1)}{RSS_e/(n-k)} = \frac{RSS_b/(k-1)}{RSS_e/(n-k)} = \frac{S_b^2}{S_e^2} \sim F_{k-1;n-k}$ $C = [F_{k-1,n-k;\alpha}, +\infty)$

· ps: 用 R 来算，lm(y ~ category1 + category2 + ..., data = ...) (ps: 如果加了-1 就不是！)

截距代表: It describes the mean effect for the {category1}.

其他代表: It is an estimate in the mean effect difference between {category1} and {category i}.

5.4 Two-way ANOVA

Target: 依照两个因素分组 (≥2) 时，不同分组之间的特征值的均值是否相等。 (假设因素一 (二) 中所有组的变量有 b(k) 个)

Definitions: Responsd Variables (k 类): $\underline{Y} = (Y_{1,1}, ..., Y_{n_1,1}, ..., Y_{1,k}, ..., Y_{n_k,k})^T$ where $Y_{i,j} \overset{i.i.d.}{\sim} N(\mu_{i,j}, \sigma^2)$ $\mu_{i,j} = \mu + \alpha_i + \beta_j$

· ps: μ: overall population mean. α_i: i^{th} 因素一对 group 的影响 β_j: j^{th} 因素二对 group 的影响

Estimators: $\hat{\mu} = \bar{y}$ $\hat{\alpha}_i = \frac{1}{k}\sum_{j=1}^{k} y_{i,j} - \bar{y}$ $\hat{\beta}_j = \frac{1}{b}\sum_{i=1}^{b} y_{i,j} - \bar{y}$ (By Least-Square Method)

$\bar{y} = \frac{1}{n}\sum_{i=1}^{b}\sum_{j=1}^{k} y_{i,j} = \frac{1}{b \cdot k}\sum_{i=1}^{b}\sum_{j=1}^{k} y_{i,j} = \frac{1}{b \cdot k}\sum_{j=1}^{k}\sum_{i=1}^{b} y_{i,j}$ $n = k \cdot b$

整体数据波动 (Overall) $RSS_y = (n-1)S_y^2 = \sum_{i=1}^{b}\sum_{j=1}^{k}(Y_{i,j} - \bar{Y})^2$

第一个因素波动 $RSS_1 = (b-1)S_1^2 = k\sum_{i=1}^{b}(\frac{1}{k}\sum_{j=1}^{k} y_{i,j} - \bar{Y})^2$

第二个因素波动 $RSS_2 = (k-1)S_2^2 = b\sum_{i=1}^{k}(\frac{1}{b}\sum_{j=1}^{b} y_{i,j} - \bar{Y})^2$

误差波动 (Unexplained) $RSS_e = (n-b-k+1)S_e^2 = \sum_{i=1}^{b}\sum_{j=1}^{k}(Y_{i,j} - \frac{1}{k}\sum_{j=1}^{k} y_{i,j} - \frac{1}{b}\sum_{j=1}^{b} y_{i,j} + \bar{Y})^2$

$RSS_y = RSS_1 + RSS_2 + RSS_e$ (我们这里忽略两个因素交错影响的问题)

Test1: 检测因素一是否对均值有影响. $H_0: \alpha_1 = \cdots = \alpha_b = 0$ $H_1: \exists \alpha_i \neq 0$

$F_1 = \frac{RSS_1/(b-1)}{RSS_e/(n-b-k+1)} = \frac{S_1^2}{S_e^2} \sim F_{b-1,n-k-b+1}$ $C = [F_{b-1,n-k-b+1;\alpha}, +\infty)$

Test2: 检测因素二是否对均值有影响. $H_0: \beta_1 = \cdots = \beta_k = 0$ $H_1: \exists \beta_j \neq 0$

$F_2 = \frac{RSS_2/(k-1)}{RSS_e/(n-b-k+1)} = \frac{S_2^2}{S_e^2} \sim F_{k-1,n-k-b+1}$ $C = [F_{k-1,n-k-b+1;\alpha}, +\infty)$

· ps: 用 R 来算，lm(y ~ T1 + T2 + ... + B1 + B2 + ... , data = ...)

截距代表: It describes the mean effect for the {T1} and {B1} combination.

其他代表: It is an estimate in the mean effect difference between {T1} and {T i} (or {B1} and {B i}).

6 Distribution Table

Distribution	PDF	Mean	Var	Other
Binomial $X \sim Bin(n,p)$	$\binom{n}{k}p^k(1-p)^{n-k}$	np	$np(1-p)$	$X, Y \sim Bin(n,p), Bin(m,p)$ $X + Y \sim Bin(n+m,p)$
For $X \sim Bin(n,p), n \to \infty, p \to 0, np \to \lambda,$ then $X \sim Poission(\lambda)$ For $X \sim Bin(n,p)$, n is large and pdf symmetric, then $X \sim N(np, np(1-p))$				
Bernoulli $X \sim Bernoulli(p)$	$\mathbb{P}(X = k) = p \; or \; 1-p$	p	$p(1-p)$	If X_i are i.i.d. $\sum X_i \sim Bin(n,p)$
Geometric $X \sim Geom(p)$	$(1-p)^{k-1}p$	$\frac{1}{p}$	$\frac{1-p}{p^2}$	$\mathbb{E}[X^2] = \frac{2}{p^2}$
Uniform $X \sim U([a,b])$	$\frac{1}{b-a}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$	$\mathbb{E}[X^2] = \frac{b^2+ab+a^2}{3}$
Exponential $X \sim Exp(\lambda)$	$\lambda e^{-\lambda x}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$	$\mathbb{E}[X^n] = \frac{n!}{\lambda^n}$
Poission $X \sim Po(\lambda)$	$e^{-\lambda} \cdot \frac{\lambda^k}{k!}$	λ	λ	$\mathbb{E}[X^2] = \lambda^2 + \lambda$
For $X, Y \sim Po(\lambda_1), Po(\lambda_2)$, then $X + Y \sim Po(\lambda_1 + \lambda_1)$				
Normal $X \sim N(\mu, \sigma^2)$	$\frac{1}{\sqrt{2\pi\sigma^2}} \cdot e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	μ	σ^2	-
For $X, Y \sim N(\mu_1, \sigma_1^2), Po(\mu_2, \sigma_2^2)$, then $aX \pm Y \pm c \sim Po(a\mu_1 \pm b\mu_2 \pm c, a^2\sigma_1^2 + b^2\sigma_2^2)$				

Beta $X \sim Beta(\alpha,\beta), \alpha,\beta > 0$	$\frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha,\beta)}, x \in [0,1]$	$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$	$B(\alpha,\beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$
Gamma $X \sim \Gamma(\alpha,\beta), \alpha,\beta > 0$	$\frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, x \geq 0$	$\frac{\alpha}{\beta}$	$\frac{\alpha}{\beta^2}$	$\Gamma(\alpha) = (\alpha-1)!$ $\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x} dx$
Chi-Squared $X \sim \chi_v^2, v > 0$	$\frac{1}{2^{\frac{v}{2}}\Gamma(\frac{v}{2})} x^{\frac{v}{2}-1} e^{-\frac{x}{2}}, x \geq 0$	v	$2v$	-
If $X_i \sim N(0,1),\ then\ \sum X_i^2 \sim \chi_n^2$ If $X \sim \chi_n^2, Y \sim \chi_m^2,\ then\ X+Y \sim \chi_{n+m}^2$				
Student's t $X \sim t_v, v > 0$	$\frac{1}{\sqrt{v}B(\frac{v}{2},\frac{1}{2})}(1+\frac{x^2}{v})^{-\frac{v+1}{2}}$	0	$\frac{v}{v-2}$	If $Z \sim N(0,1), Y \sim \chi_n^2$ $\frac{Z}{\sqrt{Y/n}} \sim t_n$
PDF Symmetric at zero and has more mass in the tails (compare to the standard normal distribution) When $n \to \infty$ it will converges to $N(0,1)$. **Cauchy Distribution**: $X \sim t_1$ and its mean and var are undefined.				
F $X \sim F(\lambda,v), v > 0$	$\frac{\lambda^{\frac{\lambda}{2}} v^{\frac{v}{2}}}{B(\frac{\lambda}{2},\frac{v}{2})} x^{\frac{\lambda}{2}-1}(\lambda x+v)^{-\frac{\lambda+v}{2}}, x \geq 0$	$\frac{v}{v-2}$	$\frac{2v^2(\lambda+v-2)}{\lambda(v-2)^2(v-4)}, v > 4$	$Y_1, Y_2 \sim \chi_{n_1}^2, \chi_{n_2}^2, X = \frac{Y_1/n_1}{Y_2/n_2}$ $X \sim F_{n_1,n_2}, X^{-1} \sim F_{n_2,n_1}$
ps: $f_{a,b;\alpha} = 1/f_{a,b;1-\alpha}$				

7 R Language

> anova(model)

	Deg.free	Res.Sum Sq.	Mean Sum Sq.	F-value	p-value
between	$k-1$	$RSS_b = RSS_y - RSS_e$	$S_b^2 = RSS_b/(k-1)$	$F = S_b^2/S_e^2$	p
within	$n-k$	RSS_e	$S_e^2 = RSS_e/(n-k)$		
Total	$n-1$	RSS_y	$S_y^2 = RSS_y/(n-1)$		

· ps: one-way ANOVA table

	Deg.free	Res.Sum Sq.	Mean Sum Sq.	F-value	p-value
category1	$b-1$	RSS_1	S_1^2	F_1	p_1
category2	$k-1$	RSS_2	S_2^2	F_2	p_1
residuals	$n-k-b+1$	RSS_e	S_e^2		
Total	$n-1$	RSS_y	S_y^2		

· ps: two-way ANOVA table   ps: R: >pnorm(p,mean,std)   ps: R: linear regression R 给的 F-statistic H_0: 除了截距其他为 0

8 Definitions and Assumptions

Population: The probability distribution that describes the frequency with which makes up all events or items for a specified research question is called the population distribution. **Population Parameter**: Typically, we are only interested in some characteristic of the population, e.g. the expectation μ, which we call the population parameter. **Sample (size)**: To investigate this, we define a sub-group of the population as sequence of random variables $X_1, ..., X_n$, where n denotes the sample size. **Representative**: If all events or items from the population has equal chance of being selected then we say that the obtained sample, $x_1, ..., x_n$, is representative of the population. **Statistic**: Any summarising transformation of $x_1, ..., x_n$ that describes the sample, e.g. the sample mean, is called a statistic. **Estimator**: A function of the random variables, $T = g(X_1, ..., X_n)$, is called an estimator if its purpose is to describe μ. **Estimate**: When the purpose of T is to describe μ, then the calculated value t is then an estimate for μ. **Sampling Distribution**: T is a function of random variables and so is random variable itself with a probability distribution called the sampling distribution.

Quantify unsertainty: 1.The uncertainty within the sample. 2.The uncertainty from the sampling.

Point Estimate: Summaries such as the sample mean, sample variance and sample median are examples of point estimates. **Sampling Variation**: As these are summaries of the available data, they are subject to Sampling Variation. **Estimate Uncertainty**: Consequently, these statistics should ideally be accompanied with an estimate of uncertainty. **Confidence Interval**: An example of this is a interval. **Confidence Level**: The width of this interval depends on the specified confidence level. We should expect that an interval to be narrower if we increase the sample size. When considering multiple representative samples from the population with the same sample size, the computed intervals will be **different**.This means that the interval **may not** contain the specific population parameter begin investigated.

Basic Distribution, Unbiased / Consistent, MLE

Assumption|Distribution, probability: Let p/θ denote the probability of {?} that ... where $X_i \sim \{?\}(p/\theta)$

Def|Unbiased: An estimator is unbiased if, on average, it equals the true value of the parameter it's estimating.

Special|$\sqrt{S^2}$: $\sqrt{S^2}$ is biased for σ at finite sample size. Because $\mathbb{E}[S] = \mathbb{E}[\sqrt{S^2}] \neq \sqrt{\mathbb{E}[S^2]} = \sqrt{\sigma^2} = \sigma$.
If S^2 is consistent in estimating σ^2, then S is Asymptotically unbiased in estimating σ.

Def|Consistent: An estimator is consistent if it converges in probability to the true value of the parameter p as the sample size increases indefinitely.

Assumption|Calculate Estimator: Let random variables sequence $X_1, ..., X_n$ represent independent $\{?: e.g. sample\}$ from a common distribution {?}, represented by random variable X, that depends on the population parameter {?}.

Assumption|Calculate Maximum Likelihood: Suppose that we have n independent observations $x_1, ..., x_n$ drawn from an {?} distribution, with probability density function $f_x = \{?\}$. Then...

Reason|$\hat{\sigma} = \sqrt{\hat{\sigma^2}}$: It's due to the invariant property of MLEs.

Reason|Uncertainty Come from: The source of uncertainty in (or in CI) an estimate comes from the principal of repeated sampling from the population.

Reason|Parameter not Random (no have uncertainty):The true {?} population parameter is assumed to be fixed (not random)

CI / PI

Conclusion|Confidence Interval: we are {?}% confident that the true {?} is between {?} and {?}
Assumption|Confidence Interval: Let $X_1, ..., X_n$ be independent and identically distributed {? : $distribution$} random variables, where {?} is unknown/known ...
Def|Confidence Interval: An interval of uncertainty for the fitted regression line, given the explantory variable x_0.
Def|Prediction Interval: An interval of uncertainty for a future observation y_0, given the explantory variable x_0.
Difference|CI and PI: PI > CI, because it's accounts fro the additional variability of the single observations, since $\sigma^2 > 0$
Reason|Why using narrower CI Interval (conver more information): The narrower CI conveys more information about {?}.
Reason|$\bar{x} = (left + right)/2$: The confidence interval for the {μ} is centred on $\bar{x}$
Reason|$\mathbb{P}(\mu < left) = \alpha/2$ is false: We cannot make such probability statement. {μ} is a fixed value (but unknown).

Linear Regression
Reason|Why not relation / one increase will lead other increase: Correlation is not Causation. A linear regression model can only describes correlated relationship.
Reason|x_i are not causal factor for Y: Regression states a correlation not a causation.
Reason|Why MLE and Least Squares Estimation are the same: The functional form of the (log-)likelihood being a sum of squared differences.
Steps|Hypothesis test:

1. Hypothesis: Let $X_1, ..., X_n$ denote ..., therefore we proposed the following Null and Alternative hypothesis statements: H_0 : $\{e.g. \mu = 0\}$ and $H_1 : \{e.g. \mu \neq 0\}$.
2. Theory: we define the test statistic {?}. At the {?} Significance level, the critical value (critical region) for Rejecting the null Hypothesis is {?}.
3. Apply: ... the observed test is {?}
4. Conclusion: $As\ z \in C\ (z \notin C)$, + Conclusion of Hypothesis test. + (可选, 就如果我们假设 A 对，但证出 B 对，问 A 没证出的原因) (i). A type II(type I) error could have occurred where the available evidence does not disagree with the null Hypothesis despite the statement being false.(ii). The ? could be ? in some other way that was not explored by this experiment.

Assumption|Power-value Calculate: Suppose that {?} has changed the {?} to now be $\{? : e.g. \mu^*\}$, then $\{? : e.g. \bar{X} \sim N(\mu^*, \sigma^2)\}$.
So the power of our hypothesis test is: ...

$$\left(e.g. \mathbb{P}(Z \in C | \mu = \mu^*) = \mathbb{P}(z < z_{1-\alpha} | \mu = \mu^*) = \mathbb{P}(z - \frac{\mu^* - \mu_0}{\sqrt{\sigma^2/n}} < z_{1-\alpha} - \frac{\mu^* - \mu_0}{\sqrt{\sigma^2/n}} | \mu = \mu^*) = \Phi(z_{1-\alpha} - \frac{\mu^* - \mu_0}{\sqrt{\sigma^2/n}})\right)$$

Assumption|Normal Linear Regression:

1. Independence between response variables.
2. Linearity of the expectation variable.
3. Have the same variance.
4. Normally distributed.
5. (可选，对于 ANOVA) The assumption can be expressed as: $Y_{i,j} \overset{i.i.d.}{\sim} N(\mu_{i,j}, \sigma^2), \mu_{i,j} = \alpha_i + \beta_j$

Steps|Determine the Linear Model: $\{e.g. Y_{i,j,k} = \alpha_i + \beta_j + \epsilon_{ijk}\}$, The $\{e.g. Y_{i,j,k}$ denote the response variables ...}, the {e.g.α_i denote the effect of ... on ...} ... ; And the {e.g.ε_{ijk} denote tghe random error associated with i,j,k ...}
Steps|Determine the Linear Model (category): Let $Y_{i,j}$ denote the {? : y} .. (见 5.3,5.4 Y 的定义). Define $\{? : category1 - group1\}$,$\{? : category2 - group1\}$ as the baseline case and denote the following binary variables: $x_{i,j,2} = \{1, if\ i = ?; 0, otherwise$ (全部都写). The linear regression model is then defines as: $\{? : e.g. Y_{i,j} = \beta_0 + \beta_1 x_{i,j,2} + +\beta_2 x_{i,j,3} + +\beta_3 z_{i,j,1} + \epsilon_{i,j}\}$ where the independent random error $\epsilon_{i,j}$ follows a normal distribution with expectation zero and common variance σ^2: {?} ~ {?} · $Y_{i,j}$ donote i^{th} observation belonging j^{th} category. $x_{i,j;r} = \mathbb{1}[\text{if } Y_{i,j}$ refers to category r]
Comment|Conclusion of the Hypothesis

1. p-value is not the probability that the null hypothesis is true. It is the probability of observing at least an extreme a value as the observed test statistic given that H_0 is true.
2. Not come from the same sample.
3. It cannot accept / make sure,...

Comment|Comment in Linear Model

1. There is a positive (negative) relationship between {?e.g.y} and {?e.g.x}
2. R-Squared value is only {?} suggesting that the model does not explain all the observed variability.
3. We could remove the $\{variable\}$ as it's p-value is too large.

Comment|Effect of the explantory variables on the response variable in model: When {x} will increases 1 unit, the {y} will increases (decreasing) {n}.